Frequency Modulation/Demodulation and Frequency Shift Keying

Josiah Germain

1 Introduction

For data to be transmitted over long distances, a high frequency is ideal. One method of turning a low frequency into a high frequency is with amplitude modulation. Amplitude modulation converts the signal carrying the information into one with a much higher frequency while maintaining the same amplitude as before. However, this method is highly susceptible to noise and has limited bandwidth. One solution to this is to carry the information in the frequency of the wave as opposed to the amplitude. This method is called Frequency Modulation (FM). FM varies the frequency of a higher frequency carrier wave in proportion to the amplitude of the signal containing the information. The equation for frequency modulation is as follows.

$$s(t) = A_c cos(2\pi f_c t + \beta sin(2\pi f_m t)) \text{ For } \beta = \frac{\Delta f}{f_m}$$

One common application of FM is Frequency Shift Keying (FSK). This is a process in which one frequency represents a digital 0 and another frequency represents a digital 1. This is a particularly useful method, as sinusoids are much better at transmitting data and then antennas can be used for wireless transmission. This also allows for modulation techniques to be used to transmit in different bandwidths.

For this experiment, a voltage controlled oscillator (VSO) will be used to generate a sine wave of different frequencies depending on the input voltage. A phase locked loop (PLL) will then be used to demodulate the FM signal. For frequency shift keying (FSK), A sequence will be generated to show how different binary values effect the frequency of the output wave.

2 Data and Analysis

To demonstrate FM, a sine wave of 4 $[V_{pp}]$ will be used as the input to the VCO. For control over the variation of the frequency at the output, the VCO will be tuned to $\pm 10[kHz]$ from the carries for an input voltage range of $\pm 2[V]$. The free standing (no input) frequency of the VCO will be set to 100 [kHz]. This high of a frequency will be hard to see on the oscilloscope so the VCO will be changed from High to Low settings. This will allow the experiment to be done at a lower frequency but the change in frequency based on the input will have similar behaviors to the high setting. The result of this is seen in Figure 1.



Figure 1: Frequency modulated signal in red with the original signal in yellow

While the change is not severe, there is a visible change in frequency of the red signal as the amplitude of the yellow signal changes. Now, to demodulate the signal to reconstruct the original information, a phase locked loop will be used. This is a circuit that used another VCO in a closed loop with the modulated signal to continuously match the frequency of the incoming wave. The result of the demodulated wave in seen in Figure 2.

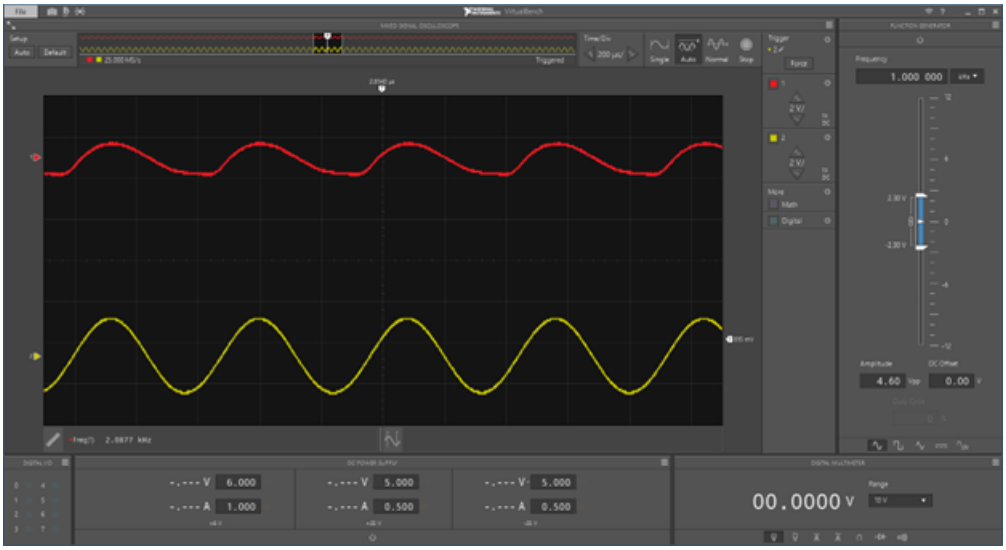


Figure 2: Original signal in yellow and demodulated signal in red

To demonstrate FSK, a 500 [Hz] sequence generator is applied to the input of the VCO. The resulting FSK signal is seen in Figure 3.

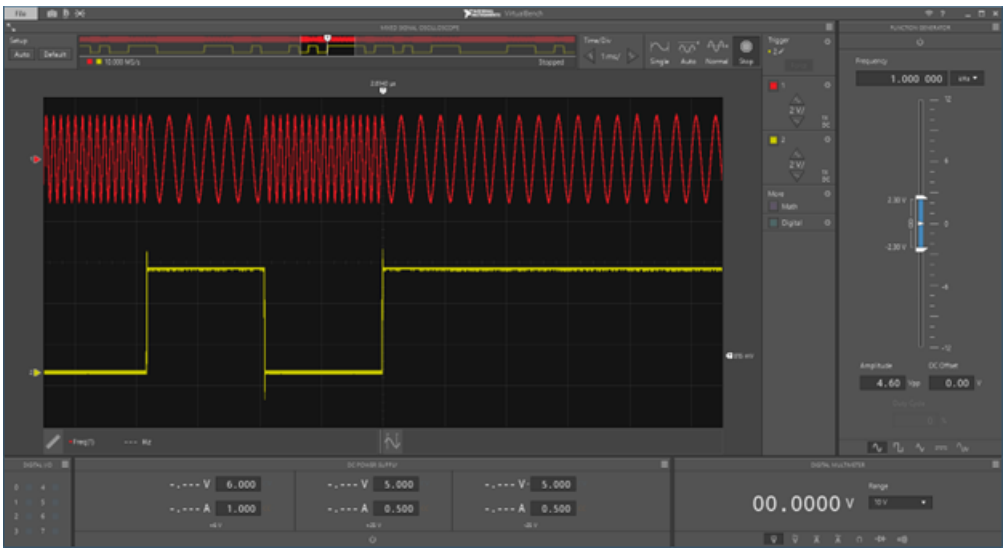


Figure 3: Sequence shown in yellow and FSK signal in red

This result clearly shows how a change in voltage directly effects the frequency of the wave generated by the VCO. Viewing this data in the time domain tells us a lot of information about this process, but the frequency domain analysis is another powerful representation of what FSK is doing. Figure 4 shows the frequency of the output wave with no input voltage,

being the free standing response of the system of 12 [kHz].

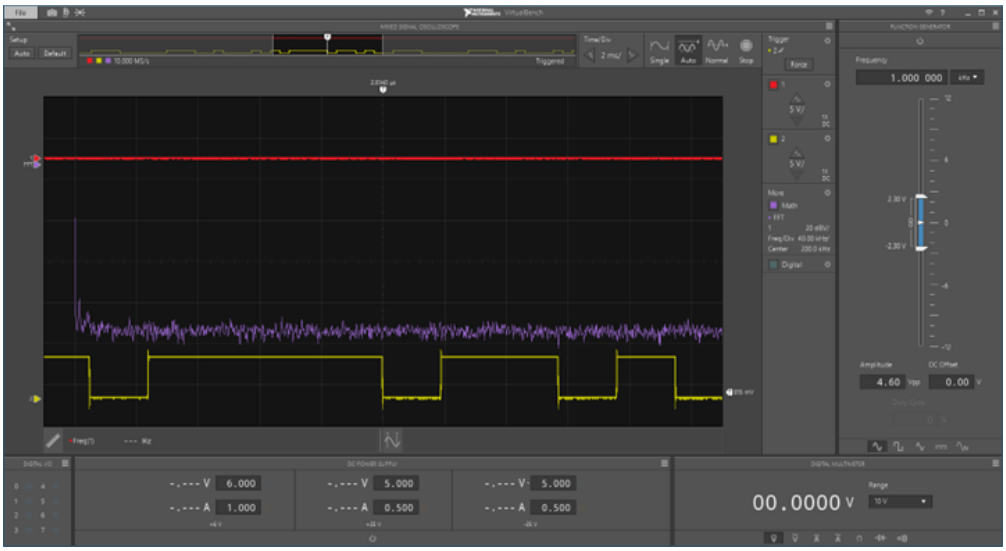


Figure 4: frequncy response of FSK system at free standing frequency of 12[kHz]

If we now include an input sequence, the frequency response will show two shifted frequencies showing the shifted frequency when the voltage is high and low. This shift will be based on the initialization done at the beginning of the lab of $\pm 3[kHz]$ from the original 12[kHz]. Figure 5 shows this new response.

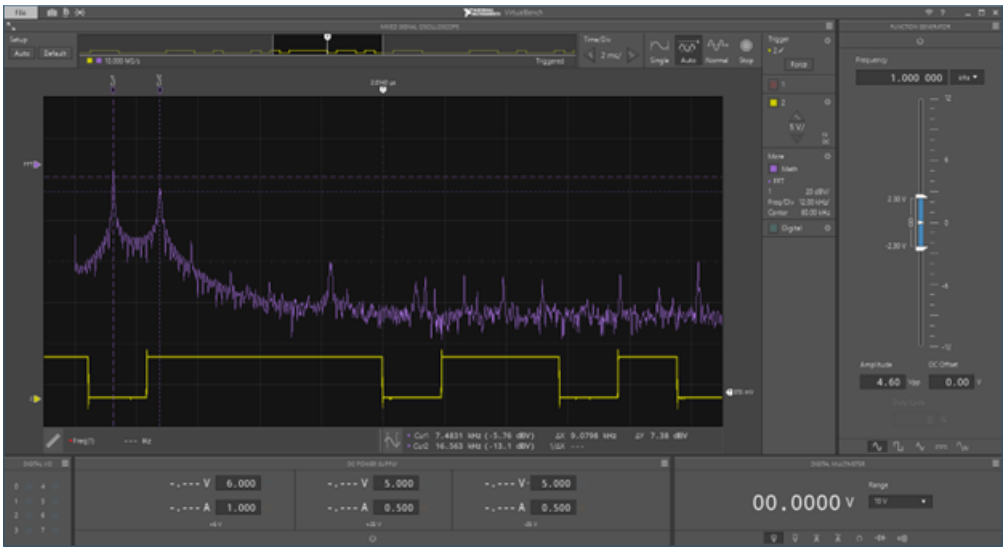


Figure 5: Frequency response of FSK system showing $\pm 3[kHz]$ change from original 12[kHz]

Certification: **I hereby certify that performance of the experiment above and the writing of this report was entirely by me (or by me and my lab partner). I understand that if this certi fication is false, I am in violation of Academic Honesty rules and may be subject to serious penalty in accordance with those rules.**

Signature Josiah Germain **Date** 11/20/25